# MOBILE COMMUNICATIONS

# UNIT 2 — CONSOLIDATED MINI-PROJECT REPORT

**Course:** CE80762: Mobile Communications

**Unit:** Wireless Communication Principles and Network Technologies

**Projects:** 2.1 Free-Space Path Loss and Modulation; 2.2 OFDM Cyclic Prefix; 2.3 Wireless Network Trade-offs

**Prepared:** 27th October 2026

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A reproducible software-based report based on the supplied September 2026 lecture notes and Unit 2 lab manual. Simulation results are computed from the accompanying Python source. Technology comparison figures are illustrative assumptions and are explicitly distinguished from published specifications.

## Executive summary

This report implements all three Unit 2 mini-project themes. Project 2.1 models ideal free-space propagation and measures uncoded BPSK, QPSK, 16-QAM and 64-QAM bit error rates in additive white Gaussian noise. Project 2.2 builds a 64-subcarrier OFDM transmitter/receiver and demonstrates the cyclic-prefix threshold under multipath. Project 2.3 compares personal-, local- and metropolitan-area wireless technology options through a transparent illustrative dataset and deployment recommendations.

## 1. Unit 2 theory reviewed

**Propagation:** reflection, diffraction and scattering create multiple propagation paths. Large-scale effects are distance-dependent path loss and shadowing; small-scale fading reflects rapid phase/amplitude changes caused by multipath. Friis is a line-of-sight baseline, whereas log-distance and Okumura-Hata models account for realistic environments.

**Information capacity:** the Shannon-Hartley bound C = B log2(1 + SNR) describes the theoretical maximum reliable rate for a specified bandwidth and linear SNR. It is not a direct formula for the BER of a particular uncoded modulation.

**Modulation and access:** BPSK, QPSK and higher-order QAM exchange robustness for bits per symbol. FDMA separates users in frequency, TDMA in time, CDMA in spreading code, and OFDMA in orthogonal subcarrier/time allocations. OFDM uses an IFFT/FFT pair and a cyclic prefix to combat multipath.

**Antennas and networks:** MIMO uses diversity, spatial multiplexing and beamforming. WPAN, WLAN, WMAN and WWAN describe increasing coverage scales; Bluetooth, Wi-Fi and WiMAX are the focus of Project 2.3.

## 2. Experimental environment and reproducibility

Python 3, NumPy, SciPy and Matplotlib. All stochastic experiments use NumPy default_rng(0). The simulation is software-only; no software-defined radio or external measurements are required. Run: `python unit2_simulations.py`. Output figures and CSV tables are generated alongside the script.

## 3. Mini Project 2.1 — Free-space path loss and digital modulation

**Objective:** visualize received power against distance and frequency, validate the 6.02 dB doubling law, and measure BER for modulation schemes over AWGN.

**Method:** Ptx = 43 dBm, Gtx = 15 dBi, Grx = 0 dBi; distances 0.05–10 km and carrier frequencies 900, 1900 and 3500 MHz. Friis equation: Prx = Ptx + Gtx + Grx - 20log10(d_km) - 20log10(f_MHz) - 32.44. For BER, use 120,000 random bits per modulation, Gray-coded mappings, unit average symbol energy, independent complex AWGN and Eb/N0 = 0, 2, ..., 16 dB.

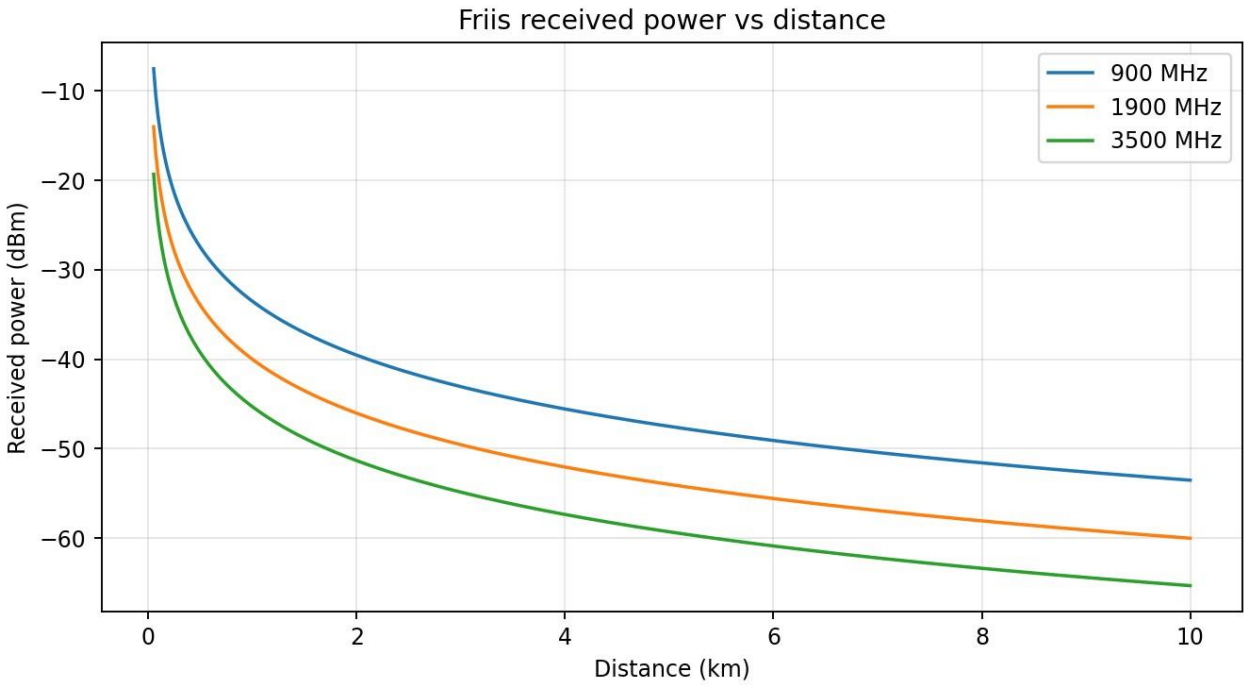



Figure 1. Friis received power falls with distance; higher carrier frequencies produce more free-space loss at equal antenna gains.

**Numerical validation:** doubling distance from 1 to 2 km reduces received power by 6.0206 dB. Doubling frequency from 900 to 1800 MHz at fixed distance also reduces received power by 6.0206 dB.

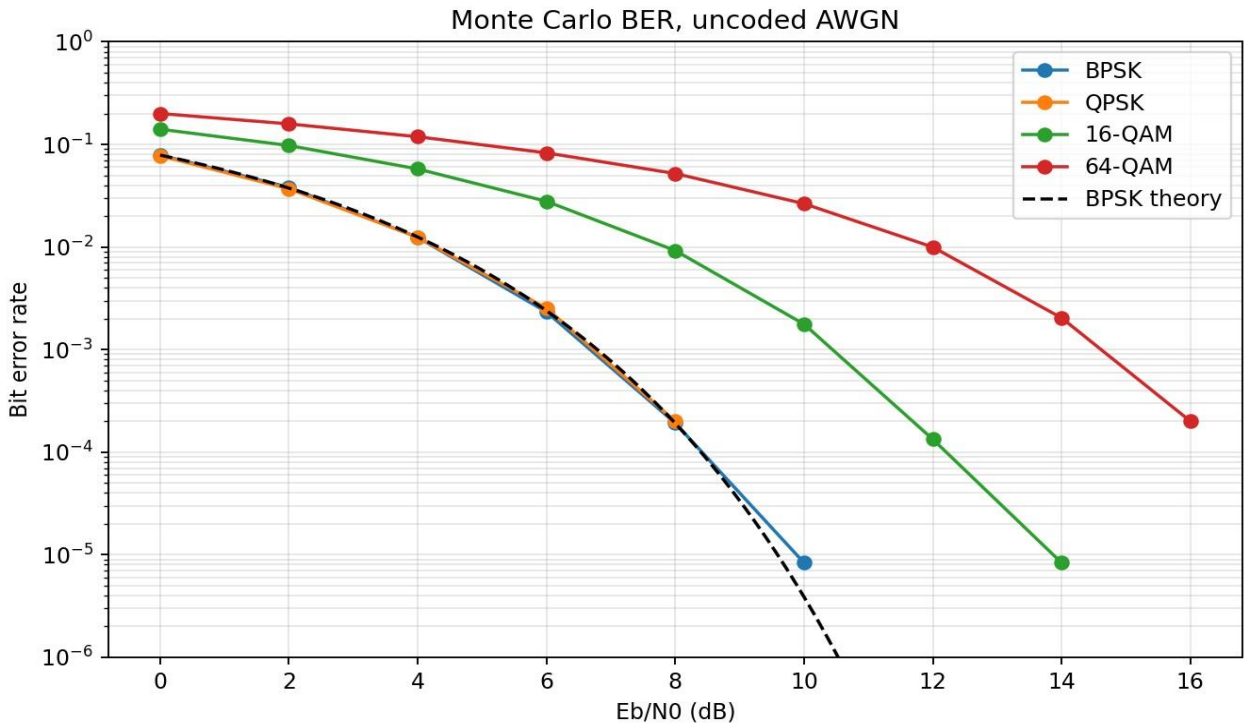



Figure 2. Simulated uncoded BER versus Eb/N0, including the theoretical BPSK reference. Missing high-SNR points indicate zero errors in the finite Monte Carlo run, not proven zero BER.

**Measured values:** at 8 dB, BPSK BER = 0.0001917, QPSK = 0.0002000, 16-QAM = 0.009225 and 64-QAM = 0.051992. For target BER = 10^-3, interpolation of the sampled log-BER curves gives approximately 6.7 dB for BPSK and 10.4 dB for 16-QAM: a gap near 3.7 dB. The gap depends on the definition of SNR, mapping and coding; the manual's approximate 8–10 dB expectation is not reproduced for uncoded Gray-coded 16-QAM at equal Eb/N0.

**Interpretation:** QPSK and BPSK have approximately equal BER versus Eb/N0 with coherent detection, although QPSK carries twice as many bits per symbol. Higher-order QAM packs constellation points more densely and therefore requires higher Eb/N0 for a target BER. Shannon-Hartley explains the overall spectral-efficiency trade-off but does not numerically predict the observed uncoded BER gap.

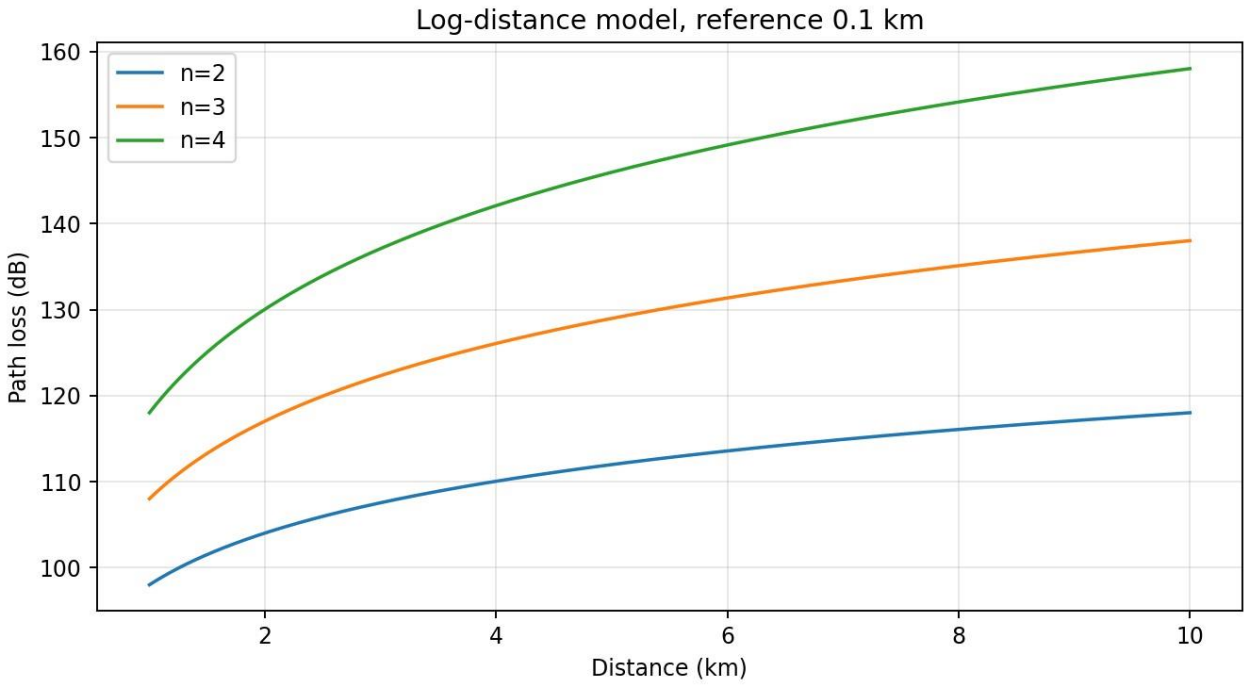



Figure 3. Optional extension: log-distance path loss for exponents n = 2, 3 and 4. Larger exponents produce faster attenuation.

**Additional lab tasks:** QPSK, 64-QAM and log-distance path loss are implemented. Rate-1/2 convolutional coding with Viterbi decoding, log-normal shadowing realizations and throughput-versus-capacity plots are not included in this implementation; these are extension tasks rather than the mandatory Project 2.1 deliverable.

## 4. Mini Project 2.2 — OFDM waveform and cyclic prefix

**Objective:** demonstrate that a cyclic prefix protects OFDM from multipath-induced inter-symbol interference (ISI) and preserves FFT-domain equalization when its length covers the channel delay spread.

**Method:** generate 500 consecutive OFDM symbols with 64 subcarriers, Gray-coded 16-QAM and IFFT modulation. Use the same deterministic 3-tap channel for every CP setting: h[0] = 1, h[3] = 0.55 exp(j0.6), h[8] = 0.4 exp(-j0.9). Thus the maximum significant delay is 8 samples. Convolve consecutive transmitted frames, discard CP, FFT, divide by the known channel frequency response and demodulate. No additive noise is applied, so any remaining errors reflect insufficient CP rather than noise. Ignore the first symbol when measuring BER to avoid start-up boundary effects.

**Self-check:** FFT(IFFT(X)) reproduces X numerically in the absence of channel impairment. CP insertion copies the tail of each time-domain symbol.

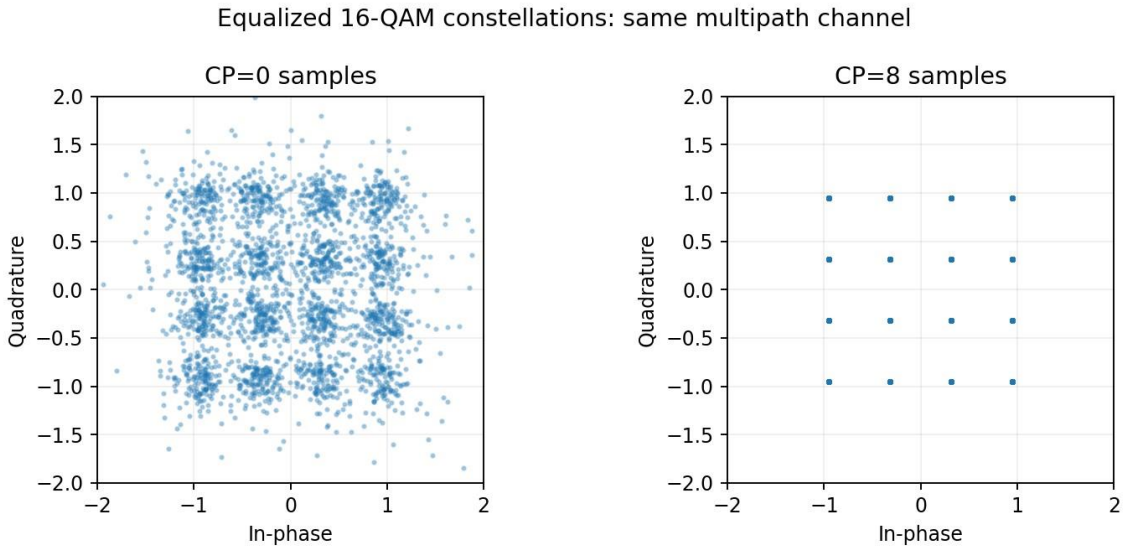



Figure 4. Equalized received 16-QAM constellations without CP (left) and with an 8-sample CP (right).

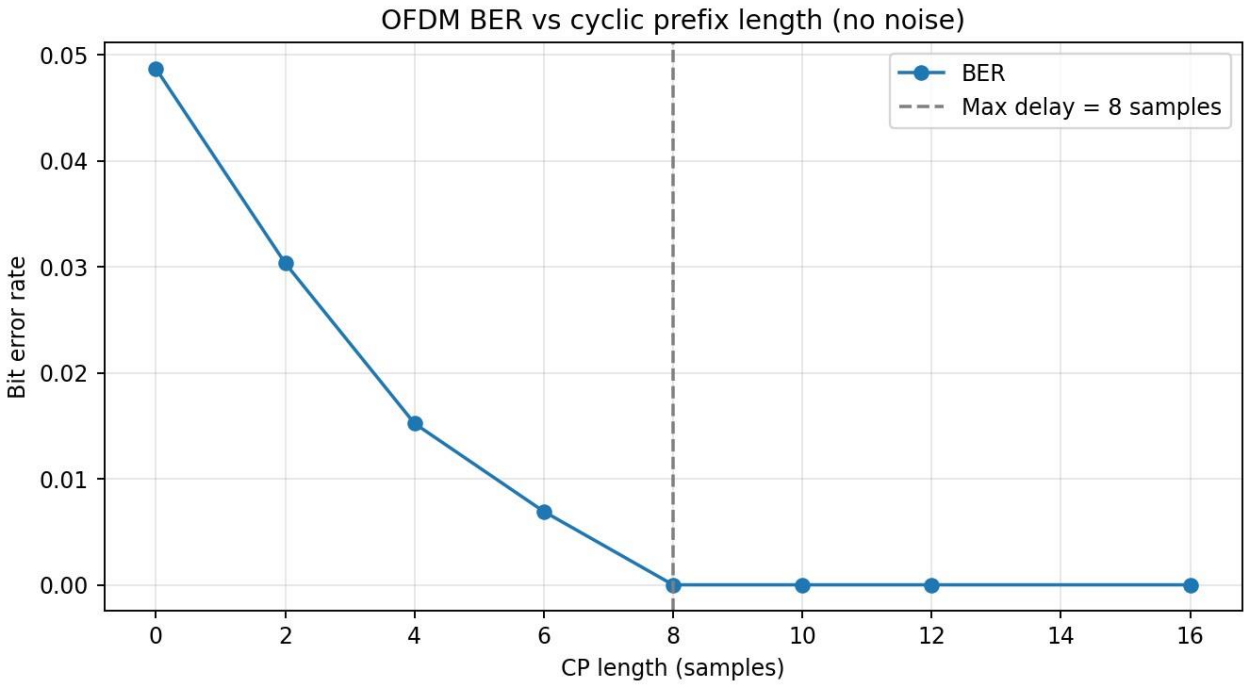



Figure 5. BER falls to zero when CP reaches the channel maximum delay of 8 samples.

| CP (samples) | BER | Mean squared error |
|---|---|---|
| 0 | 0.048715 | 0.167843 |
| 2 | 0.030350 | 0.089977 |
| 4 | 0.015226 | 0.042398 |
| 6 | 0.006920 | 0.020970 |
| 8 | 0.000000 | ~1.4e-31 |
| 10, 12, 16 | 0.000000 | ~1.4e-31 |

**Explanation:** when CP length is at least the maximum channel delay, the previous OFDM symbol's delayed tail falls within the discarded prefix. The useful FFT window then sees circular convolution, allowing one-tap

frequency-domain equalization. Shorter prefixes leave residual ISI and inter-carrier distortion. Longer prefixes do not improve this noiseless channel result but reduce useful-sample efficiency: eta = N/(N + CP), equal to 88.9% at CP = 8 and 80% at CP = 16.

## 5. Mini Project 2.3 — WPAN/WLAN/WMAN comparison

**Objective:** compare range, throughput and power requirements, and recommend wireless technologies for three deployment scenarios. The supplied Unit 2 notes specify qualitative ranges and power classes but do not provide one comparable numerical throughput figure for every technology. Accordingly, the table below uses explicit classroom scenario assumptions, NOT independently verified published performance specifications. It is suitable for illustrating the trade-off, but must be replaced with sourced device-specific measurements for a fully compliant published-specification dataset.

| Technology | Class | Assumed range | Assumed throughput | Power |
|---|---|---|---|---|
| Bluetooth BLE | WPAN | 10 m | 1 Mbps | Low |
| Wi-Fi | WLAN | 75 m | 100 Mbps | Medium |
| WiMAX | WMAN | 5000 m | 20 Mbps | High |

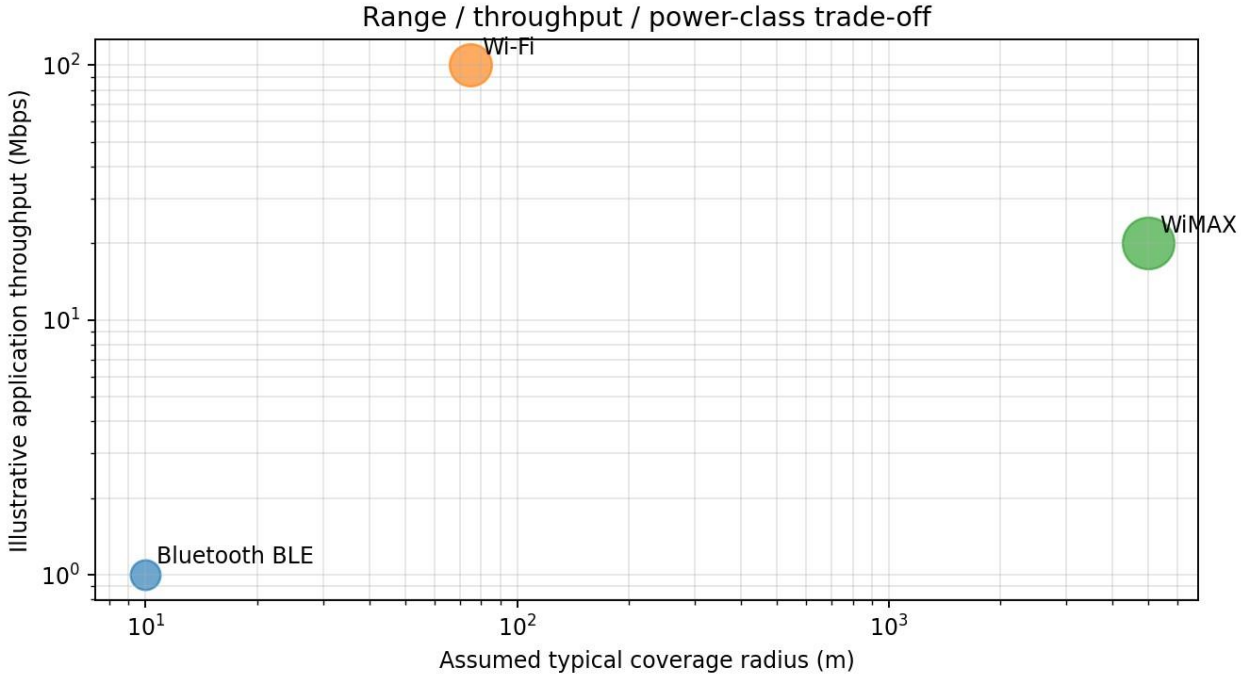



Figure 6. Illustrative range-throughput-power trade-off (marker area increases with ordinal power class). Logarithmic axes allow all three coverage scales to appear on one chart.

**Scenario A: smart home.** Requirements: low-energy sensors within roughly 10 m, low per-device throughput and battery life as the dominant constraint. **Recommendation: Bluetooth Low Energy**, with gateways where larger coverage is needed. The illustrative 1 Mbps application throughput is more than sufficient for small sensor messages, and the low power class fits battery-powered devices.

**Scenario B: campus connectivity.** Requirements: tens to hundreds of Mbps, multiple users, buildings and outdoor spaces. **Recommendation: Wi-Fi** with multiple coordinated access points rather than assuming one 75 m access point covers the campus. Its assumed 100 Mbps application throughput and established device ecosystem fit this use case.

**Scenario C: metropolitan fixed wireless link.** Requirements: connectivity over several kilometers with infrastructure power available and moderate data throughput. **Recommendation: WiMAX** as the project's

representative WMAN technology, conditional on equipment availability and local spectrum authorization. In an actual deployment, compare current LTE/5G fixed-wireless alternatives because the notes explain that LTE overtook WiMAX in adoption and ecosystem scale.

**Limitations:** nominal coverage is not guaranteed service radius; rates vary by PHY version, channel width, interference, antenna design and number of users. Power classes are qualitative rather than standardized measured watts. Equipment cost, spectrum licensing, mobility, interference and regional ecosystem support can overturn a simple numerical recommendation.

## 6. Conclusions

The Friis experiment verified the expected 6.02 dB penalty from doubling distance or carrier frequency under the stated equal-gain assumptions. BER simulations showed that BPSK and QPSK provide similar per-bit noise robustness, while higher-order QAM trades that robustness for greater bits-per-symbol capacity. The OFDM experiment demonstrated a clear CP threshold at the channel's 8-sample delay spread, with near-perfect equalization when CP was sufficient. Finally, the wireless-network comparison showed that coverage, throughput and energy requirements lead to different technology choices rather than a universal best option.

## 7. References and source integrity

[1] CE80762: Mobile Communications, Unit 2: Wireless Communication Principles and Network Technologies,lecture notes, September 2026, supplied by the student.

[2] Mobile Communications Mini Projects, Unit 2: Lab Manual, September 2026, supplied by the student.

[3] Generated numerical results: accompanying Python source and CSV outputs (October 2026).

Specification caveat: the Project 2.3 numerical dataset is explicitly illustrative and not represented as published IEEE/Bluetooth SIG performance data. The report makes no claim of hardware measurement or real-world deployment validation.